



RetailFlow

RETAIL INTELLIGENCE PLATFORM

Data Governance Policy

Governance framework for the RetailFlow Platform — designed by design,
embedded in the platform architecture.

DOCUMENT SCOPE

Governance strategy, operating model, policies and standards, GDPR alignment, consent management, quality controls, auditability, AI governance, KPIs, risk, change management and roadmap.

Official Deliverable

Data Governance

English · v1.0

RetailFlow — From customer events to trusted intelligence. Now, governed by design.

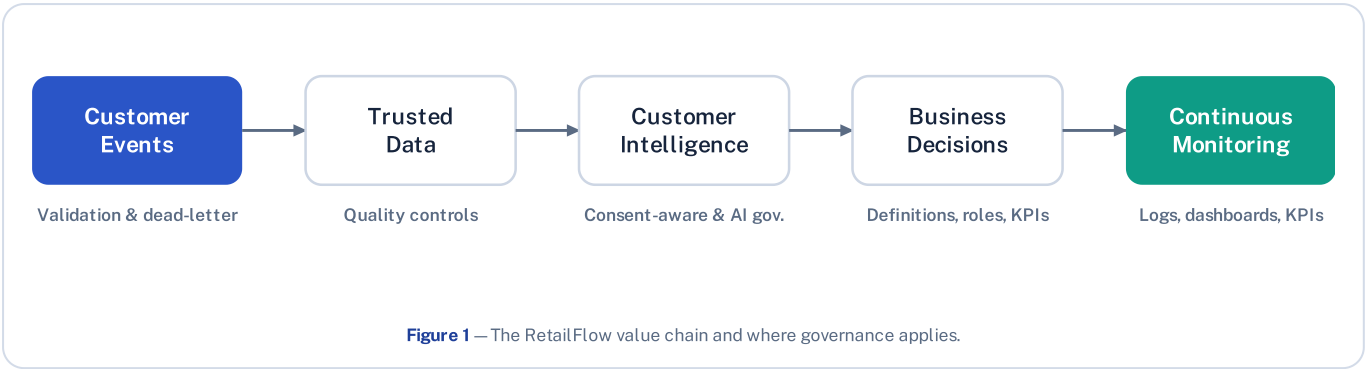
1 Organization & Data Context

RetailFlow is a recently established company that develops the RetailFlow Platform, a Retail Intelligence platform for e-commerce organizations. Because the company is new, its data governance framework was designed from the ground up rather than retrofitted onto an existing system.

The platform exists to transform customer events into trusted data, trusted data into customer intelligence, and customer intelligence into operational and strategic decision support. Governance was integrated directly into the platform — its data model, pipelines, dashboards and operational workflows — under a **Governance by Design** approach.

1.1 THE RETAILFLOW VALUE CHAIN

Each step of the value chain is kept under governance control, from raw customer events through to monitored business decisions.



1.2 DATA LANDSCAPE

The platform collects and processes customer-level data and derived AI outputs across several domains.

Data group	Representative assets
Customer	Customer profile, consent indicators, account & loyalty status, anonymization status
Behavioral events	Product views, cart events, checkout events, sessions, raw payloads
Transactions	Orders, order items, payments, returns, shipments, refunds
Product	Catalog, categories, supplier references, price, availability, metadata
Derived features	Behavioral and transactional aggregates used for analytics and ML
AI outputs	Churn predictions, CLV predictions, customer segments, drift metrics
Governance & quality	Consent records, retention policies, retention & quality logs, dead-letter events
Monitoring	Platform health, API metrics, PostgreSQL metrics, Airflow health

1.3 WHY GOVERNANCE IS CRITICAL

Because RetailFlow relies on customer-level data and AI-derived outputs, governance is a core platform requirement rather than an administrative afterthought. Without it, the platform would be exposed to several risks:

- Customer data misuse or analytics performed without consent
- Uncontrolled or indefinite data retention
- Poor data quality silently degrading AI outputs
- Unclear ownership and weak accountability
- Inability to explain or audit automated decisions
- Model-monitoring gaps and operational blind spots

2 Governance Vision, Objectives & Scope

2.1 VISION & GUIDING PRINCIPLE

GOVERNANCE VISION

RetailFlow governance ensures that customer data can be used as a trusted, compliant and valuable asset while preserving privacy, accountability, quality and auditability across the platform.

GUIDING PRINCIPLE

Data should only become intelligence if it is governed, reliable, traceable and used for a legitimate purpose.

2.2 GOVERNANCE OBJECTIVES

Objective	Intent
Clear data ownership	Every critical domain has an accountable owner for definitions, usage and quality.
Protect customer data	Apply purpose limitation, consent, minimization, storage limitation and anonymization.
Consent-aware analytics	Prioritize customers with analytics consent in customer-intelligence exploration.
Control data quality	Stop invalid events from silently entering trusted analytical tables.
Ensure auditability	Produce evidence for retention, anonymization, quality and ML actions.
Govern the AI lifecycle	Cover purpose, eligibility, explainability, monitoring, drift and oversight.
Support business adoption	Keep governance practical, role-based, measurable and visible in tools.

2.3 SCOPE

IN SCOPE

- Customer, behavioral-event, transaction and product data
- Customer features and AI outputs (churn, CLV, segments)
- Governance, quality and monitoring data

OUT OF SCOPE (FUTURE ENTERPRISE CAPABILITIES)

- Enterprise Identity & Access Management
- Single Sign-On
- Multi-region deployment
- 24/7 production support and on-call operations

The scope deliberately focuses first on the most valuable and risky domains — customer data, behavioral events and AI outputs — which most directly affect privacy, decision-making, ML reliability and compliance.

2.4 GOVERNANCE BY DESIGN & MATURITY

Principle	Implementation in RetailFlow
Privacy by Design	Consent fields, anonymization logic and retention policies in the data model.
Quality by Design	Event validation runs before event persistence.
Auditability by Design	Retention actions and quality issues are logged.
AI Governance by Design	ML predictions are monitored and linked to consent-aware exploration.
Observability by Design	Health and metrics exposed through Prometheus, Grafana and Streamlit.
Accountability by Design	Roles and personas defined for ownership, stewardship and compliance.

Maturity dimension	Current level
Data ownership	Advanced
Consent management	Advanced
Retention & anonymization	Advanced
Auditability	Advanced
Data quality	Advanced
AI governance	Intermediate–Advanced
Metadata management	Developing
Enterprise data catalog	Planned
Access management	Developing

3 Operating Model & Stakeholders

RetailFlow uses a hybrid operating model: a central governance layer defines common policies, standards and controls, while domain actors apply them in their business or technical areas.

3.1 HYBRID OPERATING MODEL

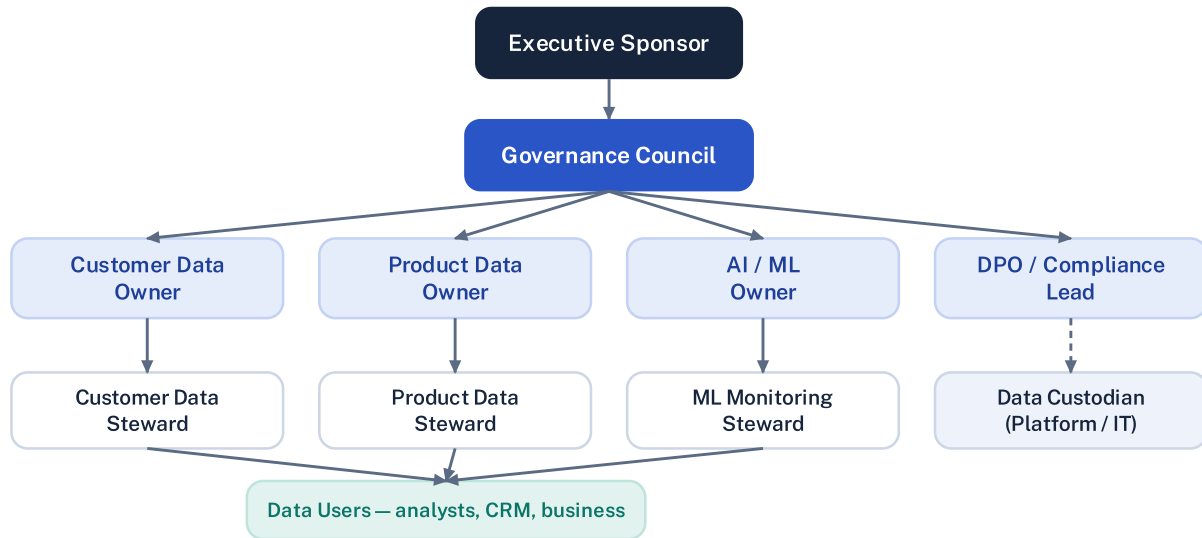


Figure 2 — Hybrid governance operating model and accountability lines.

A fully centralized model would be too rigid for a platform spanning business, engineering and AI; a fully decentralized one would create inconsistent definitions and weak accountability. The hybrid model provides central consistency, domain accountability, faster execution and clear escalation paths.

3.2 ROLES & PERSONAS

Governance role	RetailFlow persona	Main responsibility
Executive Sponsor	Chief Data & Analytics Officer	Sponsors the program and validates strategic priorities.
Governance Council	Cross-functional committee	Approves policies, reviews risks and arbitrates decisions.
Data Owner	Head of Customer Intelligence	Owns definitions, usage priorities and quality targets.
Data Steward	Senior CRM & Analytics Manager	Monitors quality, glossary, consent usage and issues.
Data Custodian	Lead Data Engineer	Operates systems, pipelines, controls and retention workflows.
DPO / Compliance Lead	Privacy & Compliance Manager	Oversees GDPR alignment, consent, retention and audit readiness.
ML Owner	Lead Machine Learning Engineer	Owns monitoring, drift, retraining and explainability.
Business Owner	Head of E-Commerce Performance	Uses intelligence for decisions and campaign prioritization.
Data Users	Marketing & CRM & business analysts	Consume governed data and report quality issues.

3.3 DECISION RIGHTS

Decision area	Owner	Consulted	Evidence required
New customer analytics use case	Data Owner	DPO, ML Owner, Business Owner	Purpose, data fields, consent requirement
New data quality rule	Data Steward	Data Custodian, Data Owner	Rule definition, severity, remediation
Retention policy update	DPO / Compliance	Data Owner, Data Custodian	Legal basis, target table, action
Model retraining approval	ML Owner	Data Owner, Data Custodian	Metrics, drift report, validation output
Governance KPI target	Governance Council	Executive Sponsor	KPI history, business impact
Access policy update	Governance Council	DPO, Data Custodian	Role mapping, security impact

3.4 GOVERNANCE CADENCE

Activity	Frequency	Responsible
Governance KPI review	Monthly	Governance Council
Data quality issue review	Weekly	Data Steward
Retention action review	Monthly	DPO / Compliance Lead
ML monitoring review	Monthly	ML Owner
Risk register review	Quarterly	Governance Council
Policy review	Quarterly	Governance Council
Accessibility & training review	Quarterly	Sponsor & Council

4 Data Domains & Classification

4.1 DATA DOMAINS UNDER GOVERNANCE

Domain	Governance focus
Customer	Privacy, consent, retention, segmentation eligibility, anonymization
Event	Validation, traceability, timeliness, dead-letter handling, quality
Transaction	Accuracy, financial consistency, reporting reliability, retention
Product	Completeness, consistency, reference values, catalog quality
AI output	Explainability, monitoring, versioning, responsible & consent-aware use

4.2 CLASSIFICATION LEVELS

Level	Description	Governance controls
Public	Shareable externally without customer risk (e.g. catalog, category labels).	Basic integrity controls
Internal	Operational/analytical data for internal use (e.g. aggregated KPIs, monitoring).	Internal access rules, documentation
Confidential	Customer-related or business-sensitive data (profiles, consent, transactions, features, ML predictions).	Consent rules, retention, audit logs, limited access

4.3 CLASSIFICATION BY DOMAIN & HANDLING

Data domain	Classification	Reason
Customer profiles, consent, behavioral events, features, ML predictions	Confidential	Customer-level or decision-influencing data
Orders & payments	Confidential	Transactional and potentially sensitive business data
Aggregated KPIs & monitoring metrics	Internal	Business / operational information kept internal
Product catalog	Public / Internal	Public info, but internal pricing or margin can be sensitive

Handling rule for Confidential data: requires a defined purpose, an appropriate role, consent consideration, a retention rule and auditability before use.

4.4 BUSINESS GLOSSARY (KEY TERMS)

Term	Definition
Analytics consent	Permission allowing customer data to be used for analytics and intelligence exploration.
Marketing consent	Permission allowing marketing activation.
Personalization consent	Permission allowing personalized recommendations or experiences.
Valid event	An event that passes required validation rules before persistence.
Dead-letter event	An event rejected by validation and isolated for review.
Churn risk	Probability or label indicating likelihood of customer disengagement.
CLV	Customer Lifetime Value — the estimated future value of a customer.
Customer segment	A business-readable group of customers with similar behavior or value.
Retention policy	Rule defining how long a data asset is kept and what action applies.
Drift	A change in data distribution that can affect model reliability.

5 Privacy, Consent & Retention

Privacy is embedded in the platform. Consent is tracked at customer level and retention is enforced through an automated, auditable workflow.

5.1 DATA POLICIES OVERVIEW

Policy	Purpose
Customer data policy	Define how customer data may be used and for which approved purposes.
Consent management policy	Control marketing, analytics and personalization usage.
Retention policy	Define how long data is retained and what action applies.
Anonymization policy	Define how customer identity data is removed or neutralized.
Data quality policy	Define validation rules and issue management.
AI governance policy	Define responsible use of ML predictions.
Access & security policy	Define access principles and technical controls.
Audit policy	Define evidence, logging and review expectations.

5.2 CONSENT MANAGEMENT

Consent field	Purpose
<code>marketing_consent</code>	Whether the customer can be targeted for marketing activation.
<code>analytics_consent</code>	Whether the customer can be used in analytics and intelligence exploration.
<code>personalization_consent</code>	Whether the customer can be used for personalization use cases.

Analytics exploration prioritizes customers with analytics consent; marketing and personalization require their respective consent. Consent must be considered before AI outputs drive customer-level actions, and consent rates (marketing, analytics, personalization, anonymized customers) are monitored on the governance dashboard.

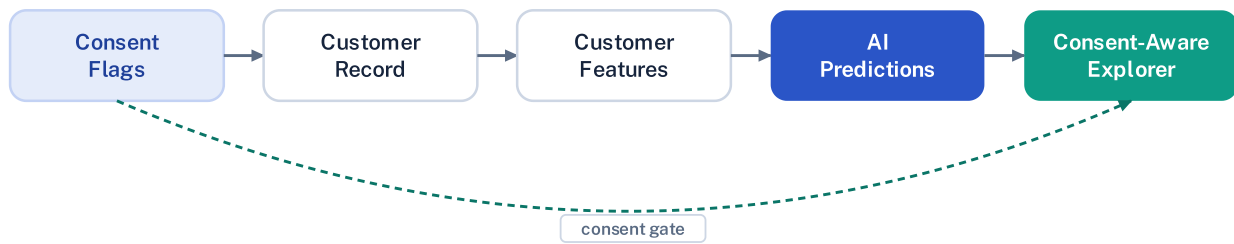


Figure 3 — Consent-aware data flow, from consent flags to governed decisions.

5.3 RETENTION & ANONYMIZATION

Retention policies are stored in `governance.data_retention_policies` and define the target domain and table, retention duration, action, owner role and legal/governance basis. Data is not retained indefinitely; personal data is anonymized when the retention condition is met, and every action is logged in `governance.retention_actions_log`.

Airflow DAG · retention_cleanup

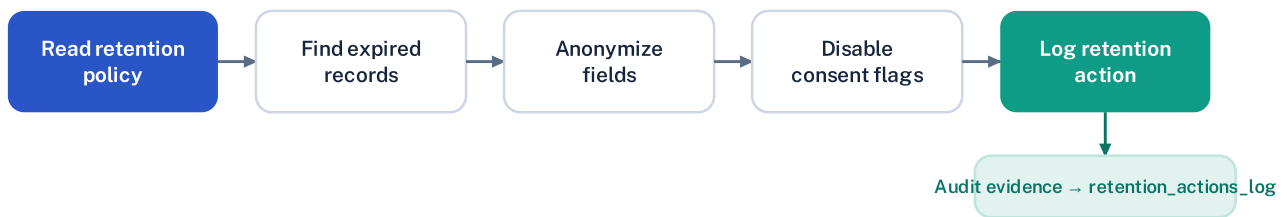


Figure 4 — Automated retention and anonymization workflow.

ANONYMIZED FIELDS

- Name, email, phone, birth date, gender
- City and postal code
- Consent flags and account status

RETENTION LOG CAPTURES

- Policy & record identifiers
- Action type and status
- Execution timestamp & component
- Action details

6 Data Quality Management

Quality controls prevent incorrect data from contaminating downstream analytics and AI outputs. Controls run inline in the real-time pipeline and are exposed through dashboards.

6.1 QUALITY DIMENSIONS & VALIDATION RULES

Dimension	Meaning in RetailFlow
Completeness	Required fields must be present.
Validity	Event types and values must be allowed.
Consistency	Events must reference existing customers and products.
Timeliness	Timestamps must be valid and usable.
Traceability	Errors must be logged and explainable.

Validation rule	Action on failure
Event identifier required	Reject invalid event
Event type allowed	Reject invalid event
Referenced customer exists	Reject invalid event
Referenced product exists	Reject invalid event
Timestamp valid	Reject invalid event

6.2 QUALITY CONTROLS

PREVENTIVE

- API request schemas
- Event validation before persistence
- Referential & event-type checks
- Timestamp validation

DETECTIVE

- Dead-letter monitoring
- Failed-rule summaries
- Scheduled Airflow quality checks
- Streamlit Data Quality page

CORRECTIVE

- Steward review of rejected (dead-letter) events
- Adjustment of quality rules where needed
- Pipeline correction by the Data Custodian
- Reprocessing path for corrected events (future improvement)

6.3 VALIDATION & DEAD-LETTER WORKFLOW

Valid events flow into trusted tables; invalid events are isolated in `governance.dead_letter_events` and logged in `governance.data_quality_logs`, then surfaced for steward review.

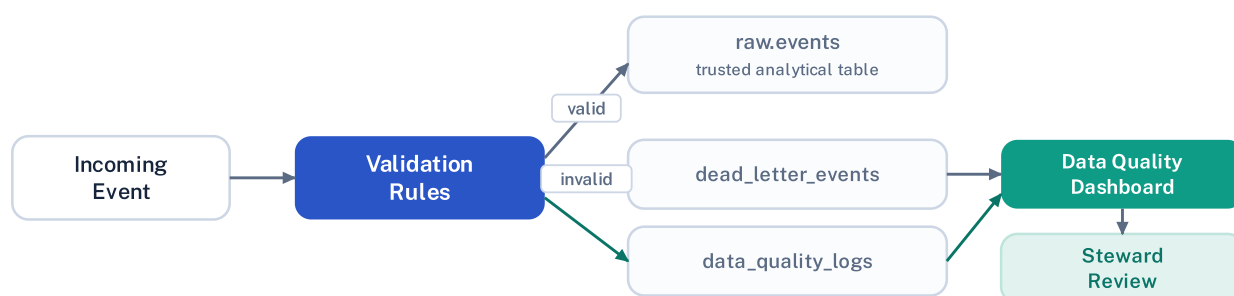


Figure 5 — Event validation, dead-letter handling and quality review.

7 Security, AI Governance & Auditability

7.1 SECURITY & ACCESS PRINCIPLES

Principle	Description
Least privilege	Users access only what they need.
Purpose-based access	Access is linked to a business purpose.
Confidentiality	Customer-level data is protected.
Auditability	Sensitive actions are traceable.
Segregation of duties	Governance decisions and technical implementation are separated.
Secure defaults	Sensitive data is not exposed by default.

Planned access improvements: role-based access control, authentication and API authorization scopes, user-level audit logging, secrets management and masking of sensitive attributes.

7.2 AI GOVERNANCE

Model	Use case	Governance concern
Churn	Identify at-risk customers	Avoid over-automated treatment; ensure explainability
CLV	Estimate customer value	Avoid unfair resource allocation without business review
Segmentation	Group customers	Ensure segments are understandable and actionable

AI predictions support — not replace — human judgment; customer-level exploration considers analytics consent; models are monitored with metrics and drift signals; outputs are explainable, versioned and timestamped. Monitored signals include churn ROC AUC, F1, precision, recall and Brier score; CLV MAE, RMSE and R²; segmentation quality; feature importance; and drift status.

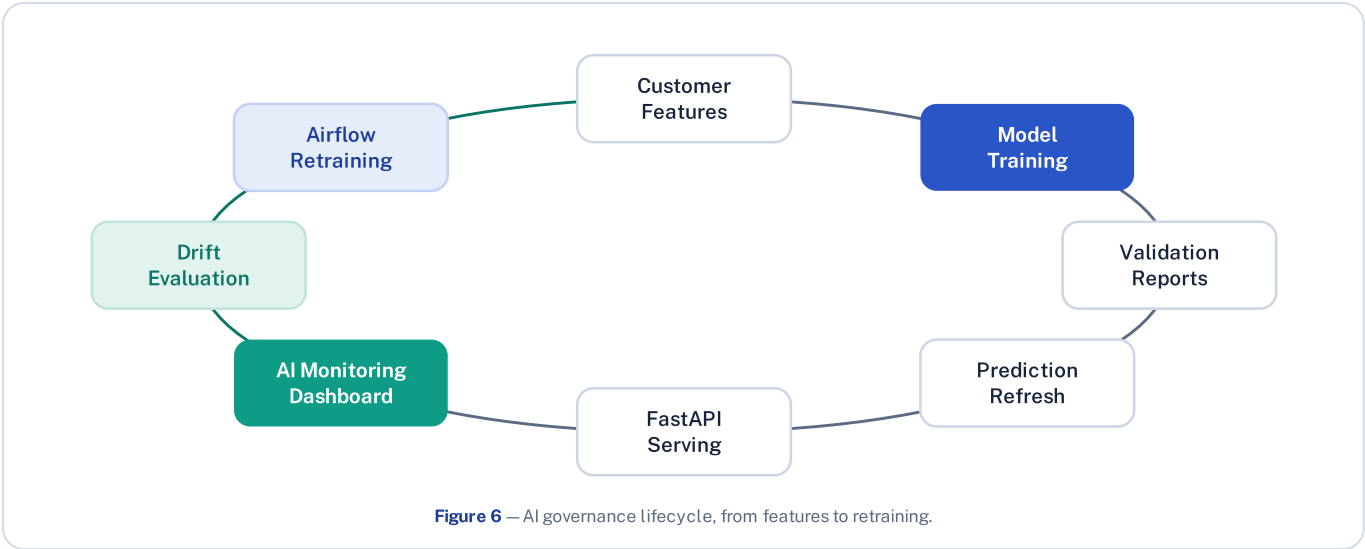


Figure 6 — AI governance lifecycle, from features to retraining.

7.3 METADATA, LINEAGE & AUDITABILITY

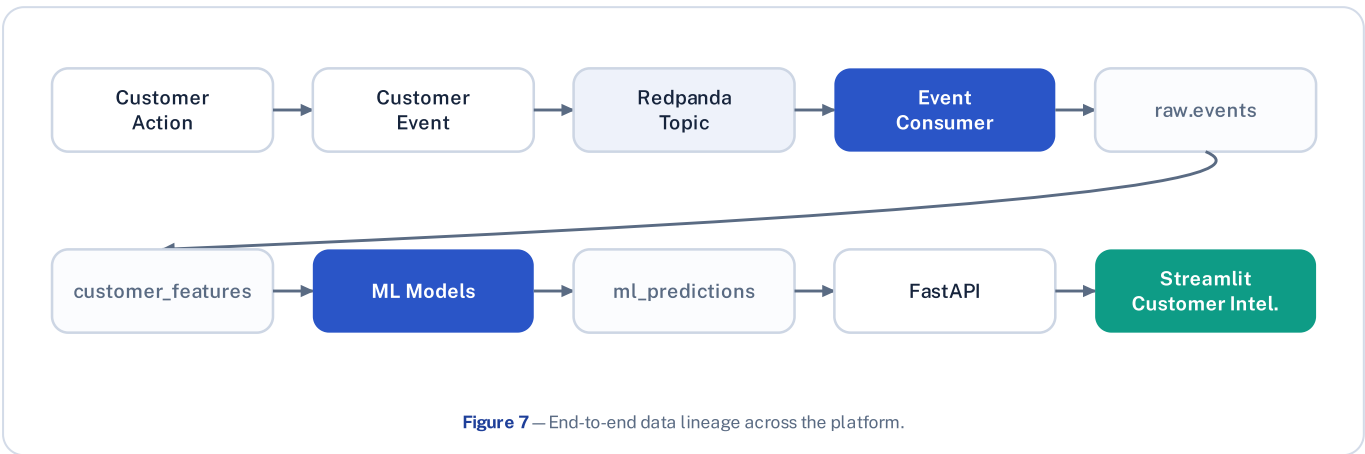


Figure 7 — End-to-end data lineage across the platform.

Audit evidence source	Proves
<code>retention_actions_log</code>	Retention and anonymization actions
<code>dead_letter_events</code>	Invalid-event isolation
<code>data_quality_logs</code>	Quality-rule execution and failures
Airflow DAG logs	Scheduled workflow execution
ML reports / Prometheus / Grafana	Model validation and platform monitoring

With this evidence, RetailFlow can answer governance questions such as which customers have analytics consent, which records were anonymized and when, which component executed an action, which events were rejected and why, whether models are monitored, and whether drift has been detected.

8 Technology, KPIs & Risk

8.1 TECHNOLOGY & TOOLING

Capability	Tool / component
Consent & governance storage	PostgreSQL customer and governance tables
Retention automation	Airflow <code>retention_cleanup</code> DAG
Event validation	Python event-consumer validators
Dead-letter & quality logs	<code>dead_letter_events</code> , <code>data_quality_logs</code>
Governance API	FastAPI <code>/governance/*</code> endpoints
Governance, Quality & AI dashboards	Streamlit pages
Operational monitoring	Prometheus and Grafana
Workflow orchestration	Airflow

8.2 GOVERNANCE KPIS

KPI	Target	Owner
Analytics consent coverage	≥ 75%	DPO / Compliance
Marketing consent coverage	≥ 50%	Business Owner
Personalization consent coverage	≥ 50%	Business Owner
Retention policy coverage (critical)	100%	DPO / Compliance
Retention action traceability	100%	Data Custodian
Data quality execution rate	≥ 95%	Data Steward
Dead-letter rate	< 2%	Data Steward
High-severity issue resolution	< 5 business days	Data Steward
ML monitoring coverage	100%	ML Owner
Governance review cadence	≥ 90%	Governance Council

8.3 RISK REGISTER

Risk	Impact	Mitigation	Owner
Personal data exposure	High	Consent, classification, access principles, anonymization	DPO
Consent misuse	High	Consent-aware dashboards and policy review	DPO
Retention failure	High	Retention policies, Airflow cleanup, audit logs	Data Custodian
Poor data quality	High	Validation rules, quality logs, dead-letter handling	Data Steward
ML drift	Medium	Drift monitoring and retraining DAG	ML Owner
Unclear ownership	Medium	Operating model and role mapping	Council
Dashboard misuse	Medium	Training, metric guides, human oversight	Business Owner
Accessibility gap	Medium	Accessible, multi-format training materials	Sponsor

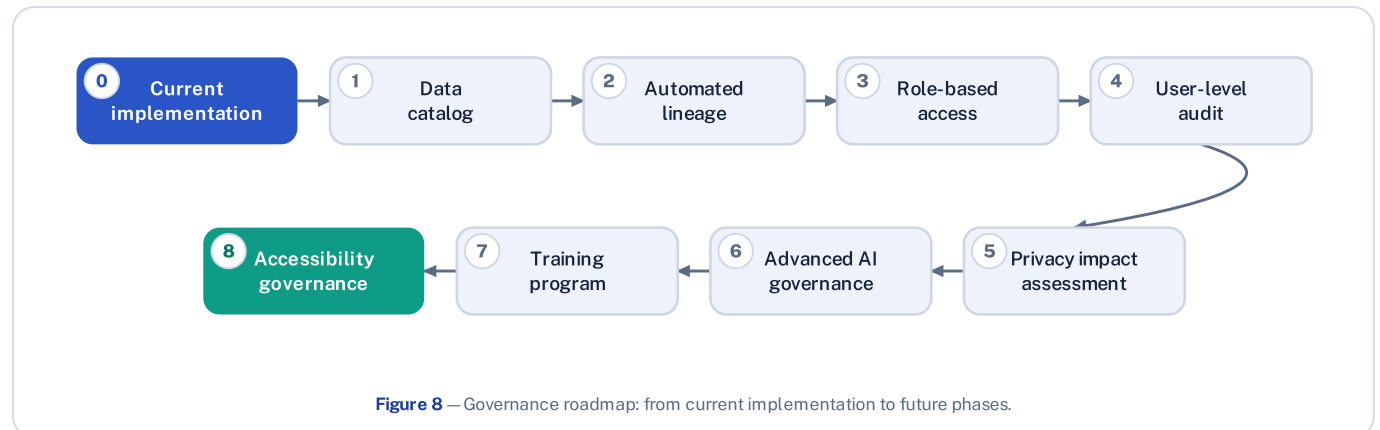
9 Roadmap & Change Management

RetailFlow already includes substantial governance implementation. The roadmap therefore covers both completed work and future phases, supported by a change-management plan.

9.1 IMPLEMENTATION STATUS (COMPLETED)

Area	Completed implementation
Consent management	Customer consent fields and consent dashboard
Analytics consent	Consent-aware customer explorer
Retention & anonymization	Retention policy table, retention_cleanup DAG, anonymization logic, audit log
Data quality	Validation rules, quality logs, dead-letter handling
Governance & quality dashboards	Streamlit Data Governance and Data Quality pages
AI governance	AI monitoring, drift and explainability views
Orchestration	Airflow governance and ML workflows

9.2 FUTURE ROADMAP



Phase	Improvement	Purpose
1	Formal data catalog	Improve discoverability and ownership visibility
2	Automated lineage	Trace data from source events to dashboards and ML outputs
3	Role-based access control	Strengthen access governance by persona
4	User-level audit logging	Track access to sensitive datasets and dashboards
5	Privacy impact assessment	Formalize privacy review for new use cases
6	Extended AI governance	Add fairness checks, model registry and approval workflow
7	Governance training program	Institutionalize awareness and adoption
8	Accessibility governance	Keep training and documentation inclusive

9.3 CHANGE MANAGEMENT & TRAINING

Governance succeeds only if people adopt it. The plan explains why governance matters, clarifies responsibilities, trains data users, reduces resistance and creates feedback loops.

Training module	Audience	Format
Governance fundamentals	All data users	Short online module
Consent & privacy	Marketing, analytics, business	Workshop + reference sheet
Data quality issue reporting	Data users & stewards	Practical session
AI output interpretation	Business & ML users	Dashboard walkthrough
Retention & anonymization	DPO, custodians, stewards	Process training

Adoption strategy: start with the customer and event domains, train the most active users first, make governance visible through dashboards, review KPIs monthly, collect feedback and extend governance to further domains.

9.4 INCLUSION & ACCESSIBILITY

Governance only works if users can understand and apply it. Training and documentation should avoid unnecessary jargon, be available in multiple languages where teams are international, and support screen-reader-compatible documents, captions, clear contrast, readable fonts,

non-visual alternatives for diagrams and simplified summaries for non-specialist audiences. Inclusion is integrated into the change-management plan rather than treated as an afterthought.

10 Conclusion

The RetailFlow data governance framework is an operational component of the platform, not a separate theoretical layer. It covers ownership, policies, consent, retention, anonymization, quality, AI monitoring, auditability, inclusion and continuous improvement.

Its main strength is that governance is embedded directly in the platform design — visible through consent-aware customer intelligence, retention and anonymization workflows, dead-letter handling, quality logs, governance APIs, dashboards, Airflow workflows and AI monitoring.

RetailFlow therefore demonstrates a mature approach to data governance for a modern Retail Intelligence platform: customer data is more trustworthy, business intelligence more reliable, AI outputs more responsible, and platform operations more auditable.

IN ONE LINE

From customer events to trusted intelligence — governed, by design.